



DATA ANALYSIS AND VISUALIZATION

INSTRUCTOR: UMME AMMARAH





NATURAL LANGUAGE PROCESSING (NLP)

TEXT CLASSIFICATION



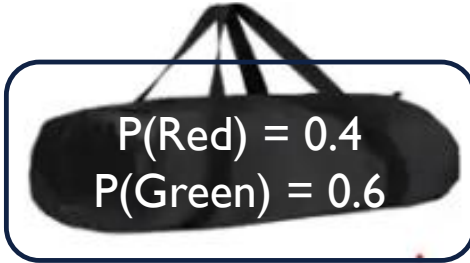
PROBABILITIES (RECAP)

- $P(X = 0) =$
- $P(Y = 3) =$
- $P(X = 1, Y = 2) =$
- $P(Y = 2, X = 1) =$
- $P(X = 1 \mid Y = 2) =$
- $P(Y = 2 \mid X = 1) =$

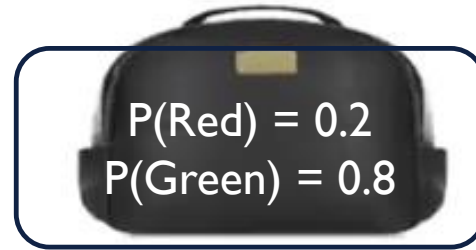
X	Y
0	0
0	1
1	0
1	2
2	3
2	0
2	3
1	3
1	2
0	3
0	2
0	0

BAYES RULE

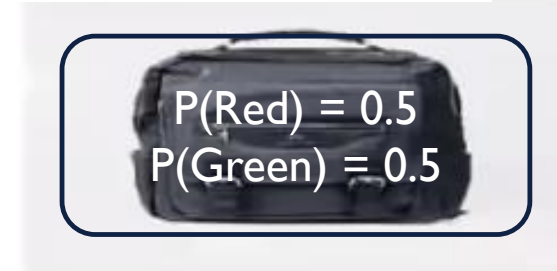
$$P(\text{bag1}) = 0.3$$



$$P(\text{bag2}) = 0.5$$



$$P(\text{bag3}) = 0.2$$



- $P(\text{bag2}) =$
- $P(\text{Red} | \text{bag1}) =$
- $P(\text{Red}) =$
- $P(\text{bag1} | \text{Red}) =$

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

CONT...

- $P(A, B)$ or $P(A \cap B) = ?$
- $P(A, B) = P(A | B) * P(B) = P(B | A) * P(A)$
- $P(A, B | C) = P(A | B, C) * P(B | C) = P(B | A, C) * P(A)$
- $P(A \cap B \cap C) = P(A, B, C)$
 $= P(A|B,C) * P(B,C) = P(A|B,C) * P(B|C) * P(C)$

- $P(A \cap B \cap C \cap D) = ?$

CONT.

if all events are independent

- $P(A, B) = P(A) * P(B)$
- $P(A, B, C) =$
- $P(A, B, C, D) =$
- $P(A_1, A_2, \dots, A_n) =$

Conditionally independent

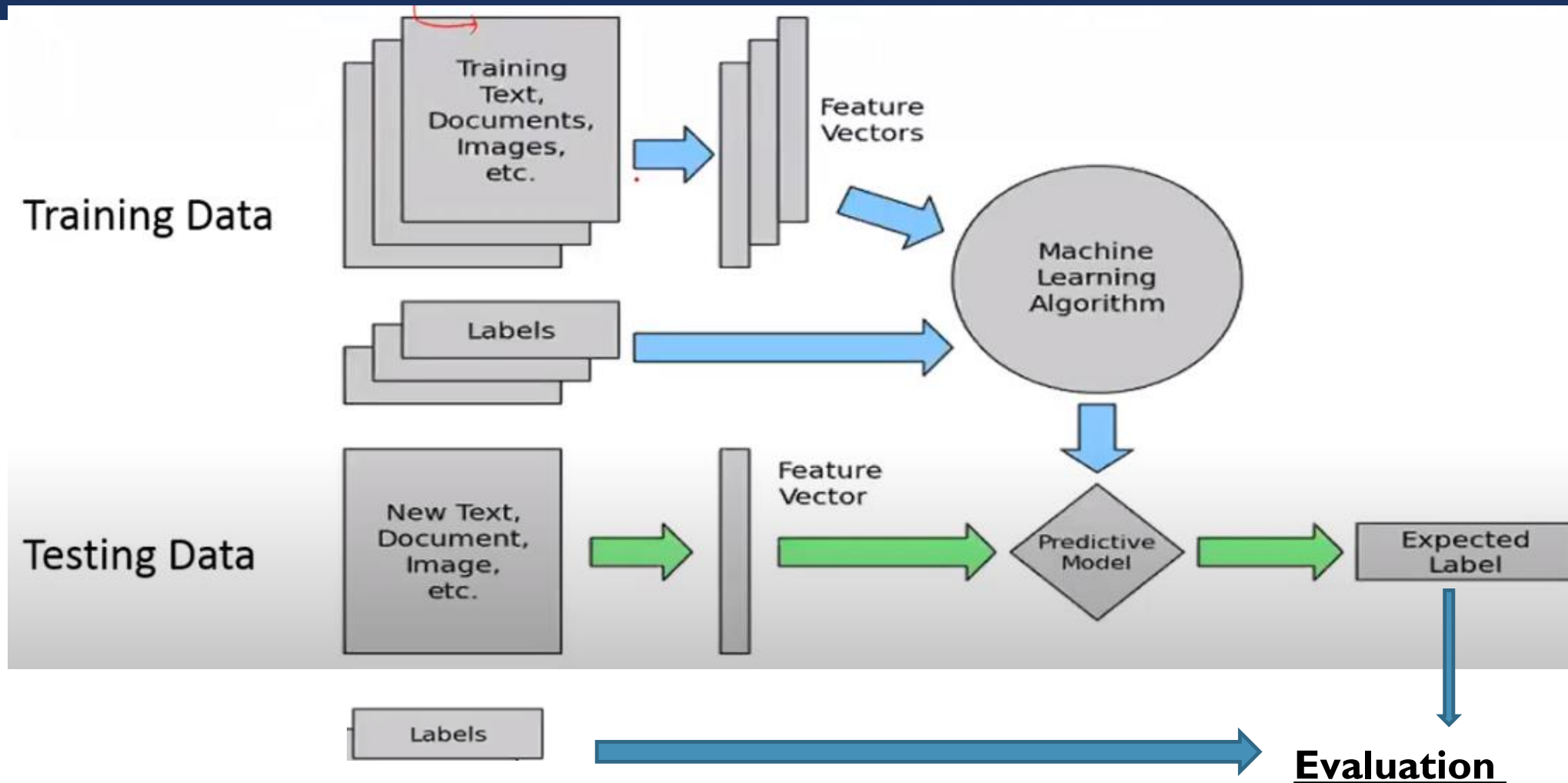
- $P(A, B | C) = P(A | C) * P(B | C)$
- $P(A, B, C | D) =$
- $P(A_1, A_2, \dots, A_n | Z) =$



NAÏVE BAYES CLASSIFIER



NAÏVE BAYES CLASSIFIER



ASSUMPTION

- Conditional independence: Assume the feature probabilities x are independent given the class c .

$$P(x | c)$$

$$P(x_1, x_2, \dots, x_n | c)$$

$$P(x_1, \dots, x_n | c) = P(x_1 | c) \bullet P(x_2 | c) \bullet P(x_3 | c) \bullet \dots \bullet P(x_n | c)$$

NAÏVE BAYES

The diagram shows the Naïve Bayes formula with four labels and arrows pointing to its components:

- Likelihood**: Points to $P(x | c)$ in the numerator.
- Class Prior Probability**: Points to $P(c)$ in the numerator.
- Posterior Probability**: Points to $P(c | x)$ on the left side of the equation.
- Predictor Prior Probability**: Points to $P(x)$ in the denominator.

$$P(c | x) = \frac{P(x | c) P(c)}{P(x)}$$

$$P(c | X) = P(x_1 | c) \times P(x_2 | c) \times \cdots \times P(x_n | c) \times P(c)$$

NAIVE BAYES CLASSIFIER

$$c_{MAP} = \operatorname{argmax}_{\hat{c} \in C} P(c \mid d)$$

MAP is “maximum a posteriori” = most likely class

$$= \operatorname{argmax}_{c \in C} \frac{P(d \mid c)P(c)}{P(d)}$$

Bayes Rule

$$= \operatorname{argmax}_{\hat{c} \in C} P(d \mid c)P(c)$$

Dropping the denominator

EXAMPLE

• Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

• Testing Data

1, 1, 0, 0, 0, 1 y
 1, 0, 1, 1, 1, 1 y

$$P(y) = 1/2$$

$$P(n) = 1/2$$

$$P(x_1=1|y) = 2/3$$

$$P(x_1=0|y) = 1/3$$

$$P(x_1=1|n) = 1/3$$

$$P(x_1=0|n) = 2/3$$

$$P(x_2=1|y) = 2/3$$

$$P(x_2=0|y) = 1/3$$

$$P(x_2=1|n) = 2/3$$

$$P(x_2=0|n) = 1/3$$

$$P(x_3=1|y) = 0$$

$$P(x_3=0|y) = 1$$

$$P(x_3=1|n) = 1/3$$

$$P(x_3=0|n) = 2/3$$

$$P(x_4=1|y) = 2/3$$

$$P(x_4=0|y) = 1/3$$

$$P(x_4=1|n) = 2/3$$

$$P(x_4=0|n) = 1/3$$

$$P(x_5=1|y) = 2/3$$

$$P(x_5=0|y) = 1/3$$

$$P(x_5=1|n) = 2/3$$

$$P(x_5=0|n) = 1/3$$

$$P(x_6=1|y) = 2/3$$

$$P(x_6=0|y) = 1/3$$

$$P(x_6=1|n) = 1/3$$

$$P(x_6=0|n) = 2/3$$

SMOOTHING

- A solution would be **Laplace smoothing** , which is a technique for smoothing categorical data.
- A small-sample correction, or **pseudo-count**, will be incorporated in every probability estimate.
- Consequently, no probability will be zero.
- This is a way of regularizing Naive Bayes, and when the pseudo-count is zero, it is called Laplace smoothing.
- While in the general case it is often called **Lidstone smoothing**.

• Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

$$P(y) = 1/2$$

$$P(n) = 1/2$$

$$P(x1=1|y) = 3/5$$

$$P(x1=0|y) = 2/5$$

$$P(x1=1|n) = 2/5$$

$$P(x1=0|n) = 3/5$$

$$P(x2=1|y) = 3/5$$

$$P(x2=0|y) = 2/5$$

$$P(x2=1|n) = 3/5$$

$$P(x2=0|n) = 2/5$$

$$P(x3=1|y) = 1/5$$

$$P(x3=0|y) = 4/5$$

$$P(x3=1|n) = 2/5$$

$$P(x3=0|n) = 3/5$$

$$P(x4=1|y) = 3/5$$

$$P(x4=0|y) = 2/5$$

$$P(x4=1|n) = 3/5$$

$$P(x4=0|n) = 2/5$$

$$P(x5=1|y) = 3/5$$

$$P(x5=0|y) = 2/5$$

$$P(x5=1|n) = 3/5$$

$$P(x5=0|n) = 2/5$$

$$P(x6=1|y) = 3/5$$

$$P(x6=0|y) = 2/5$$

$$P(x6=1|n) = 2/5$$

$$P(x6=0|n) = 3/5$$

• Testing Data

1, 1, 0, 0, 0, 1 y

1, 0, 1, 1, 1, 1 y

$$\operatorname{argmax}_{c \in C} P(x|c)P(c)$$



TEXT CLASSIFICATION



IS THIS SPAM?

Subject: Important notice!

From: Stanford University <newsforum@stanford.edu>

Date: October 28, 2011 12:34:16 PM PDT

To: undisclosed-recipients;;

Greats News!

You can now access the latest news by using the link below to login to Stanford University News Forum.

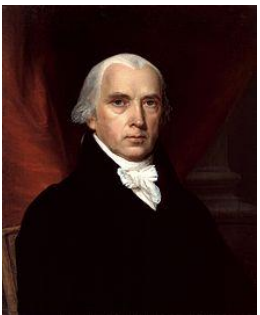
<http://www.123contactform.com/contact-form-StanfordNew1-236335.html>

Click on the above link to login for more information about this new exciting forum. You can also copy the above link to your browser bar and login for more information about the new services.

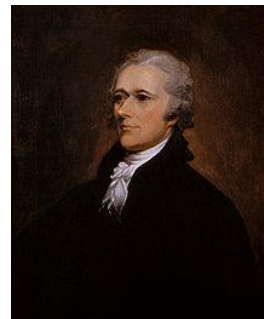
© Stanford University. All Rights Reserved.

WHO WROTE WHICH FEDERALIST PAPERS?

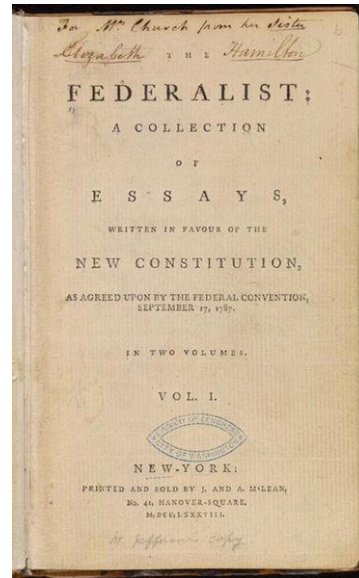
- 1787-8: anonymous essays try to convince New York to ratify U.S Constitution: Jay, Madison, Hamilton.
- Authorship of 12 of the letters in dispute
- 1963: solved by Mosteller and Wallace using Bayesian methods



James Madison



Alexander Hamilton



WHAT IS THE SUBJECT OF THIS MEDICAL ARTICLE?

MEDLINE Article



MeSH Subject Category Hierarchy

- Antagonists and Inhibitors
- Blood Supply
- Chemistry
- Drug Therapy
- Embryology
- Epidemiology
- ...

POSITIVE OR NEGATIVE MOVIE REVIEW?

- + *...zany characters and richly applied satire, and some great plot twists*
- *It was pathetic. The worst part about it was the boxing scenes...*
- + *...awesome caramel sauce and sweet toasty almonds. I love this place!*
- *...awful pizza and ridiculously overpriced...*

POSITIVE OR NEGATIVE MOVIE REVIEW?

- + ...zany characters and **richly** applied satire, and some **great** plot twists
- It was **pathetic**. The **worst** part about it was the boxing scenes...
- + ...**awesome** caramel sauce and sweet toasty almonds. I **love** this place!
- ...**awful** pizza and **ridiculously** overpriced...

WHY SENTIMENT ANALYSIS?

- *Movie*: is this review positive or negative?
- *Products*: what do people think about the new iPhone?
- *Public sentiment*: how is consumer confidence?
- *Politics*: what do people think about this candidate or issue?
- *Prediction*: predict election outcomes or market trends from sentiment

SUMMARY: TEXT CLASSIFICATION

- Sentiment analysis
- Spam detection
- Authorship identification
- Language Identification
- Assigning subject categories, topics, or genres...

TEXT CLASSIFICATION: DEFINITION

- *Input:*

- a document d
- a fixed set of classes $C = \{c_1, c_2, \dots, c_J\}$

- *Output:* a predicted class $c \in C$

CLASSIFICATION METHODS: HAND-CODED RULES

- Rules based on combinations of words or other features
 - spam: black-list-address OR (“dollars” AND “you have been selected”)
- Accuracy can be high
 - If rules carefully refined by expert
- But building and maintaining these rules is expensive

CLASSIFICATION METHODS: SUPERVISED MACHINE LEARNING

- *Input:*

- a document d
- a fixed set of classes $C = \{c_1, c_2, \dots, c_J\}$
- A training set of m hand-labeled documents
 $(d_1, c_1), \dots, (d_m, c_m)$

- *Output:*

- a learned classifier $\gamma: d \rightarrow c$

CLASSIFICATION METHODS: SUPERVISED MACHINE LEARNING

- Any kind of classifier
 - Naïve Bayes
 - Logistic regression
 - Neural networks
 - k-Nearest Neighbors
 - ...



NAÏVE BAYES FOR TEXT CLASSIFICATION

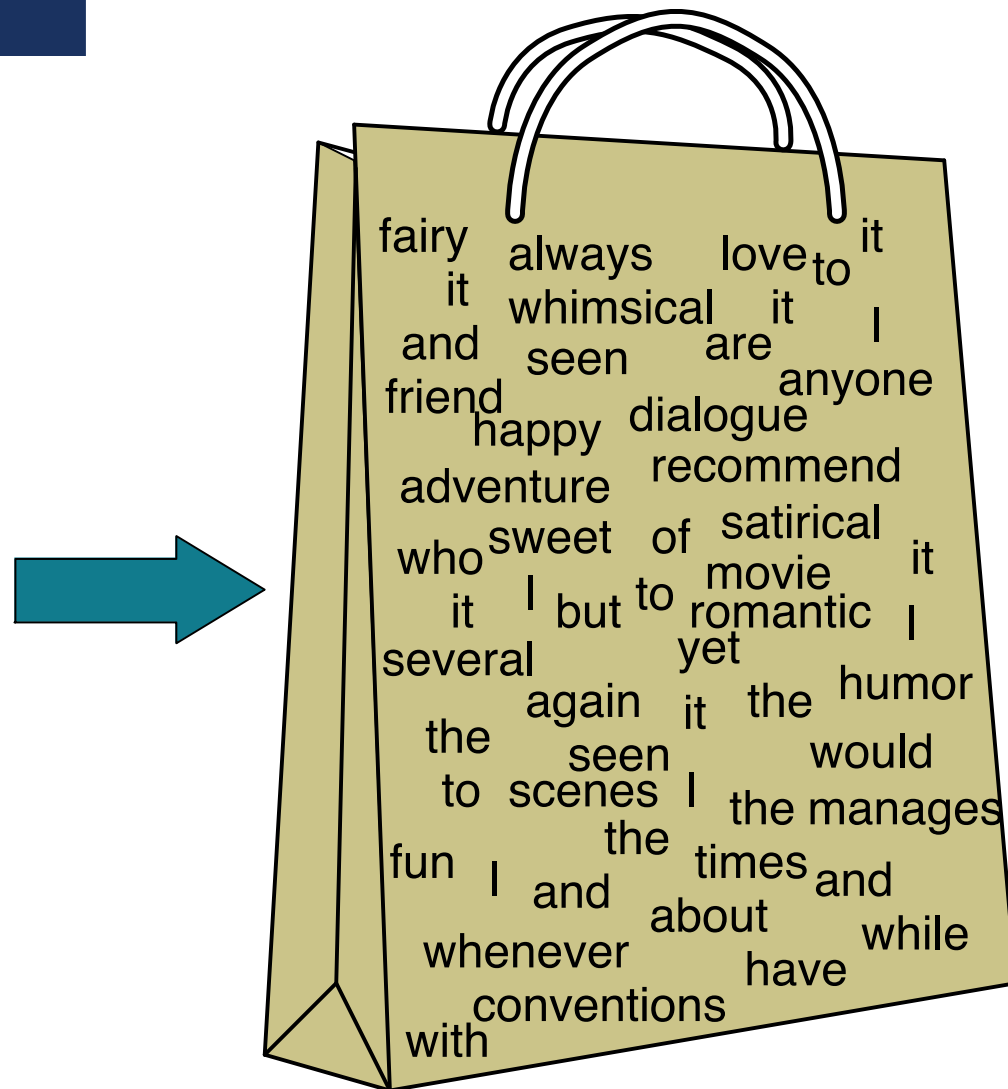


NAIVE BAYES INTUITION

- Simple ("naive") classification method based on Bayes rule
- Relies on very simple representation of document
 - **Bag of words**

THE BAG OF WORDS REPRESENTATION

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

THE BAG OF WORDS REPRESENTATION

Y (

seen	2
sweet	1
whimsical	1
recommend	1
happy	1
...	...

)

= C



BAYES' RULE APPLIED TO DOCUMENTS AND CLASSES

- For a document d and a class c

$$P(c | d) = \frac{P(d | c)P(c)}{P(d)}$$

NAIVE BAYES CLASSIFIER

"Likelihood"

"Prior"

$$c_{MAP} = \operatorname{argmax}_{c \in C} P(d \mid c)P(c)$$

$$= \operatorname{argmax}_{c \in C} P(x_1, x_2, \dots, x_n \mid c)P(c)$$

Document d
represented as
features $x_1 \dots x_n$

NAÏVE BAYES CLASSIFIER

$$C_{MAP} = \operatorname{argmax}_{c \in C} P(x_1, x_2, \dots, x_n \mid c) P(c)$$

$O(|X|^n \cdot |C|)$ parameters

Could only be estimated if a very, very large number of training examples was available.

How often does this class occur?

We can just count the relative frequencies in a corpus

MULTINOMIAL NAIVE BAYES INDEPENDENCE ASSUMPTIONS

$$P(x_1, x_2, \dots, x_n \mid c)$$

- **Bag of Words assumption:** Assume position doesn't matter
- **Conditional Independence:** Assume the feature probabilities $P(x_i \mid c_j)$ are independent given the class c .

$$P(x_1, \dots, x_n \mid c) = P(x_1 \mid c) \bullet P(x_2 \mid c) \bullet P(x_3 \mid c) \bullet \dots \bullet P(x_n \mid c)$$

MULTINOMIAL NAIVE BAYES CLASSIFIER

$$c_{MAP} = \operatorname{argmax}_{c \in C} P(x_1, x_2, \dots, x_n | c) P(c)$$

$$c_{NB} = \operatorname{argmax}_{\hat{c} \in C} P(c_j) \tilde{\mathbf{O}}_{x \hat{c}} P(x | c)$$

APPLYING MULTINOMIAL NAIVE BAYES CLASSIFIERS TO TEXT CLASSIFICATION

positions $\leftarrow$ all word positions in test document

$$c_{NB} = \underset{c_j \in C}{\operatorname{argmax}} P(c_j) \prod_{i \in \text{positions}} P(x_i | c_j)$$

PROBLEMS WITH MULTIPLYING LOTS OF PROBS

- There's a problem with this:

$$c_{NB} = \underset{c_j \in C}{\operatorname{argmax}} P(c_j) \prod_{i \in \text{positions}} P(x_i | c_j)$$

- Multiplying lots of probabilities can result in floating-point underflow!
- $.0006 * .0007 * .0009 * .01 * .5 * .000008....$
- Idea: Use logs, because $\log(ab) = \log(a) + \log(b)$
- We'll sum logs of probabilities instead of multiplying probabilities!

WE ACTUALLY DO EVERYTHING IN LOG SPACE

Instead of this:

$$c_{NB} = \operatorname{argmax}_{c_j \in C} P(c_j) \prod_{i \in \text{positions}} P(x_i | c_j)$$

This:

$$c_{NB} = \operatorname{argmax}_{c_j \in C} \left[\log P(c_j) + \sum_{i \in \text{positions}} \log P(x_i | c_j) \right]$$

Notes:

- 1) Taking log doesn't change the ranking of classes!
The class with highest probability also has highest log probability!
- 2) It's a linear model:
Just a max of a sum of weights: a **linear** function of the inputs
So naive bayes is a **linear classifier**

LEARNING THE MULTINOMIAL NAIVE BAYES MODEL

- First attempt: maximum likelihood estimates
 - simply use the frequencies in the data

$$\hat{P}(c_j) = \frac{N_{c_j}}{N_{total}}$$

$$\hat{P}(w_i | c_j) = \frac{\text{count}(w_i, c_j)}{\sum_{w \in V} \text{count}(w, c_j)}$$

PARAMETER ESTIMATION

$$\hat{P}(w_i | c_j) = \frac{\text{count}(w_i, c_j)}{\sum_{w \in V} \text{count}(w, c_j)}$$

fraction of times word w_i appears
among all words in documents of
topic c_j

- Create mega-document for topic j by concatenating all docs in this topic
- Use frequency of w in mega-document

PROBLEM WITH MAXIMUM LIKELIHOOD

- What if we have seen no training documents with the word ***fantastic*** and classified in the topic **positive** (***thumbs-up***)?

$$\hat{P}(\text{"fantastic"} \mid \text{positive}) = \frac{\text{count}(\text{"fantastic"}, \text{positive})}{\sum_{w \in V} \text{count}(w, \text{positive})} = 0$$

- Zero probabilities cannot be conditioned away, no matter the other evidence!

$$c_{MAP} = \operatorname{argmax}_c \hat{P}(c) \prod_i \hat{P}(x_i \mid c)$$

LAPLACE (ADD-1) SMOOTHING FOR NAÏVE BAYES

$$\hat{P}(w_i | c) = \frac{\text{count}(w_i, c) + 1}{\sum_{w \in V} (\text{count}(w, c) + 1)}$$

$$= \frac{\text{count}(w_i, c) + 1}{\left(\sum_{w \in V} \text{count}(w, c) \right) + |V|}$$

MULTINOMIAL NAÏVE BAYES: LEARNING

- From training corpus, extract *Vocabulary*

- Calculate $P(c_j)$ terms

- For each c_j in C do

$docs_j \leftarrow$ all docs with class = c_j

$$P(c_j) \leftarrow \frac{|docs_j|}{|\text{total \# documents}|}$$

- Calculate $P(w_k | c_j)$ terms

- $Text_j \leftarrow$ single doc containing all $docs_j$

- For each word w_k in *Vocabulary*

$n_k \leftarrow$ # of occurrences of w_k in $Text_j$

$$P(w_k | c_j) \propto \frac{n_k + a}{n + a |Vocabulary|}$$

UNKNOWN WORDS

- What about unknown words
 - that appear in our test data
 - but not in our training data or vocabulary?
- We **ignore** them
 - Remove them from the test document!
 - Pretend they weren't there!
 - Don't include any probability for them at all!
- Why don't we build an unknown word model?
 - It doesn't help: knowing which class has more unknown words is not generally helpful!

STOP WORDS

- Some systems ignore stop words
 - **Stop words:** very frequent words like *the* and *a*.
 - Sort the vocabulary by word frequency in training set
 - Call the top 10 or 50 words the **stopword list**.
 - Remove all stop words from both training and test sets
 - As if they were never there!
- But removing stop words doesn't usually help
 - So in practice most NB algorithms use **all** words and **don't** use stopwords lists

LET'S DO A WORKED SENTIMENT EXAMPLE!

	Cat	Documents
Training	- - - + +	just plain boring entirely predictable and lacks energy no surprises and very few laughs very powerful the most fun film of the summer
Test	?	predictable with no fun

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer
Test	?	predictable with no fun

1. Prior from training:

$$\hat{P}(c_j) = \frac{N_{c_j}}{N_{total}}$$

$$P(-) = 3/5$$

$$P(+) = 2/5$$

2. Drop "with"

3. Likelihoods from training:

$$p(w_i|c) = \frac{\text{count}(w_i, c) + 1}{(\sum_{w \in V} \text{count}(w, c)) + |V|}$$

$$P(\text{"predictable"}|-) = \frac{1+1}{14+20} \quad P(\text{"predictable"}|+) = \frac{0+1}{9+20}$$

$$P(\text{"no"}|-) = \frac{1+1}{14+20} \quad P(\text{"no"}|+) = \frac{0+1}{9+20}$$

$$P(\text{"fun"}|-) = \frac{0+1}{14+20} \quad P(\text{"fun"}|+) = \frac{1+1}{9+20}$$

4. Scoring the test set:

$$P(-)P(S|-) = \frac{3}{5} \times \frac{2 \times 2 \times 1}{34^3} = 6.1 \times 10^{-5}$$

$$P(+)P(S|+) = \frac{2}{5} \times \frac{1 \times 1 \times 2}{29^3} = 3.2 \times 10^{-5}$$



EVALUATION



EVALUATION

- Let's consider just binary text classification tasks
- Imagine you're the CEO of Delicious Pie Company
- You want to know what people are saying about your pies
- So you build a "Delicious Pie" tweet detector
 - Positive class: tweets about Delicious Pie Co
 - Negative class: all other tweets

THE 2-BY-2 CONFUSION MATRIX

gold standard labels

		gold positive	gold negative	
<i>system output labels</i>	system positive	true positive	false positive	precision = $\frac{tp}{tp+fp}$
	system negative	false negative	true negative	
		recall = $\frac{tp}{tp+fn}$		accuracy = $\frac{tp+tn}{tp+fp+tn+fn}$

EVALUATION: ACCURACY

- Why don't we use **accuracy** as our metric?
- Imagine we saw 1 million tweets
 - 100 of them talked about Delicious Pie Co.
 - 999,900 talked about something else
- We could build a dumb classifier that just labels every tweet "not about pie"
 - It would get 99.99% accuracy!!! Wow!!!!
 - But useless! Doesn't return the comments we are looking for!
 - That's why we use **precision** and **recall** instead

EVALUATION: PRECISION

- % of items the system detected (i.e., items the system labeled as positive) that are in fact positive (according to the human gold labels)

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

EVALUATION: RECALL

- % of items actually present in the input that were correctly identified by the system.

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

WHY PRECISION AND RECALL

- Our dumb pie-classifier
 - Just label nothing as "about pie"

Accuracy=99.99%

but

Recall = 0

- (it doesn't get any of the 100 Pie tweets)

Precision and recall, unlike accuracy, emphasize true positives:

- finding the things that we are supposed to be looking for.

A COMBINED MEASURE: F

- F measure: a single number that combines P and R:

$$F_{\beta} = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

- We almost always use balanced F_1 (i.e., $\beta = 1$)

$$F_1 = \frac{2PR}{P + R}$$